

Algebraic Topology 3

Labix

February 9, 2026

Abstract

Algebraic Topology 3 picks up from Algebraic Topology 2 and defines the final invariant for homotopy equivalence called the homotopy groups. We shall see that such homotopy groups is a complete invariant for CW-complexes up to homotopy equivalence. CW-complexes also benefit from the homotopy groups with the homotopy analogue of excision and a unique new theorem called the suspension theorem that implies stability of the homotopy groups.

References:

- Notes on Algebraic Topology by Oscar Randal-Williams:
The first chapter gives a complete treatment of the first three sections of these notes, as well as providing the importance of fibrations on the higher homotopy groups. These notes are highly recommended to understanding the first three sections.
- Algebraic Topology by Allen Hatcher:
A more or less complete dictionary on all topics of these notes. However it is prone to the same problem in the sense that Hatcher's book is rather terse and definitions and parts of some theorems are scattered throughout the paragraphs rather than having a complete statement for reference. Nevertheless it is still the standard reference of the notes, albeit organized in a slightly different way.
- A non-visual proof that higher homotopy groups are abelian by Shintaro Fushida-Hardy:
This short piece of article proves that the higher homotopy groups are abelian in a purely algebraic way. Most geometric visualization of such a proof has the same underlying idea as the algebraic method.

Contents

1	The Higher Homotopy Groups	3
1.1	The n th Homotopy Groups	3
1.2	Functorial Properties of Homotopy	6
1.3	The Action of the Fundamental Group	8
1.4	Relative Homotopy Groups	8
1.5	Induced Maps of Relative Homotopy Groups	8
1.6	Long Exact Sequence in Homotopy Groups	9
1.7	Homotopical Connectivity	10
2	Weak Equivalences and CW-Complexes	11
2.1	Weak Homotopy Equivalence	11
2.2	Whitehead's Theorem	11
2.3	Cellular Approximations	11
2.4	CW Approximations	13
3	Main Results of Homotopy Theory on CW-Complexes	14
3.1	Excision for Homotopy Groups	14
3.2	Hurewicz's Theorem	14
3.3	Eilenberg-MacLane Spaces	15
4	Generalized (Co)Homology Theories	16
4.1	Weak Equivalences and (Co)Homological Isomorphisms	16
4.2	Generalized Homology Theories	16
4.3	Generalized Cohomology Theories	16
5	The Stable Phenomena	18
5.1	Freudenthal Suspension Theorem	18
5.2	The Stable Homotopy Groups	19
6	Bonus?	21
6.1	Postnikov Towers	21
7	Kunneth Theorem for Homology	22
7.1	The Universal Coefficient Theorem for Homology	22
7.2	Kunneth Theorem for Homology	22
8	Spectral Sequences in Algebraic Topology	23
8.1	Spectral Sequences in Topology	23
9	H-Spaces	25
9.1	H-Spaces and the Hopf Algebra Structure on Cohomology	25
9.2	Homotopy Associative H-Space and the Pontryagin Product	25
9.3	Group Like H-Spaces	26
10	The Free Monoid of a Space	27
10.1	The James Construction	27
10.2	The Infinite Symmetric Product	27
11	The Super Lie Algebra Structure in Homotopy Groups	29
11.1	The Samelson Product	29
11.2	The Whitehead Product	29
11.3	The Hilton Milnor Theorem	30

1 The Higher Homotopy Groups

The journey of Algebraic Topology began with the fundamental group, where we assigned a group to every space functorially. The notion of fundamental group heavily involves the notion of homotopy and therefore is heavily related to the notion of homotopy. However, one realizes that even with Seifert-van Kampen theorem and the theory of covering spaces, it is not easy to compute the fundamental group of a space. This is partly, but not wholly due to the fundamental group is in general not abelian. If we instead work in an abelian setting, one is able to distinguish two non-isomorphic groups simply by analysing the torsion subgroups. Therefore we refine the concept of the fundamental group and procured the notion of homology and cohomology. Both functorial invariants now produce graded abelian groups for each space, one for each dimension $n \in \mathbb{N}$. In the case of cohomology, there is a canonical ring structure on cohomology that interacts with the topology of the underlying space.

Now we turn to the final main invariant of topological spaces. The homotopy groups $\pi_n(X, x_0)$ serves as both a generalization of the fundamental group $\pi_1(X, x_0)$ in higher dimensions and a homotopic analogue to homology via the Hurewicz homomorphism

$$h : \pi_n(X) \rightarrow H_n(X)$$

It is a strong invariant that is closely related to the notion of homotopy, all the while having mostly abelian groups as its output. The trade off is that the homotopy groups are very hard to compute. Such trade off has led to the blossoming of Algebraic Topology in its fullest. For instance, stable homotopy theory stems from a crucial fact called the Freudenthal suspension theorem, which states that such a sequence

$$\pi_n(X) \rightarrow \pi_{n+1}(\Sigma X) \rightarrow \cdots$$

eventually stabilizes for large enough n .

In this chapter we will closely study the n th homotopy groups such as its properties and develop tools to compute them.

1.1 The n th Homotopy Groups

We begin not with the definition of the homotopy groups, but rather a slight generalization of pointed spaces and maps between them.

Definition 1.1.1 (Pairs of Space) Let X be a topological space. A pair of space is a pair (X, A) where $A \subseteq X$ is a subspace of X . A map of pairs $f : (X, A) \rightarrow (Y, B)$ is a continuous map $f : X \rightarrow Y$ such that $f(A) \subseteq B$.

Definition 1.1.2 (Homotopy between Maps of Pairs) Let $f, g : (X, A) \rightarrow (Y, B)$ be maps of pairs. A homotopy between f and g is a homotopy $H : X \times [0, 1] \rightarrow Y$ such that $H(A \times [0, 1]) \subseteq B$.

Definition 1.1.3 (The n th Homotopy Groups) Let (X, x_0) be a pointed space. Define the n th homotopy group $\pi_n(X, x_0)$ to be

$$\pi_n(X, x_0) = \frac{\left\{ \gamma : (I^n, \partial I^n) \rightarrow (X, \{x_0\}) \mid \gamma \text{ is continuous} \right\}}{\simeq}$$

where we say that $f \simeq g$ if there exists a homotopy between f and g .

Notice that the definition coincides with that of the fundamental group when $n = 1$, and hence π_n is indeed a generalization.

Now we have defined the set $\pi_n(X, x_0)$ for a pointed space to have the word group in its name. We will also need to procure a canonical group structure on the set $\pi_n(X, x_0)$. This will be similar with that of the fundamental group.

Definition 1.1.4 (Concatenation) Let $n \geq 1$. Let (X, x_0) be a pointed space. Let $f, g : (I^n, \partial I^n) \rightarrow (X, x_0)$ be maps. Define the composition of f and g by the formula

$$(f \cdot g)(t_1, \dots, t_n) = \begin{cases} f(2t_1, t_2, \dots, t_n) & \text{if } 0 \leq t_1 \leq \frac{1}{2} \\ g(2t_1 - 1, t_2, \dots, t_n) & \text{if } \frac{1}{2} \leq t_1 \leq 1 \end{cases}$$

for $f, g \in \pi_n(X, x_0)$.

Notice that concatenation is really just the same concatenation between elements of the fundamental group but instead with more coordinates. The group structure on $\pi_n(X, x_0)$ uses concatenation and such a proof also uses the same homotopies as in Algebraic Topology 1, but with more coordinates.

Theorem 1.1.5 Let (X, x_0) be a pointed space and $n \geq 1$. The operation $\cdot$ on the equivalence classes in $\pi_n(X, x_0)$ is well defined and endows it with the structure of a group.

Proof We first show that the operation is well defined on $\pi_n(X, x_0)$. Suppose that $f_1 \stackrel{\partial}{\simeq} g_1 : (I^n, \partial I^n) \rightarrow (X, x_0)$ via the homotopy H_1 and $f_2 \stackrel{\partial}{\simeq} g_2 : (I^n, \partial I^n) \rightarrow (X, x_0)$ via the homotopy H_2 . Consider the map $H : I^n \times [0, 1] \rightarrow X$ defined by

$$H(x_1, \dots, x_n, t) = \begin{cases} H_1(2x_1, \dots, x_n, t) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \\ H_2(2x_1 - 1, \dots, x_n, t) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \end{cases}$$

Now when $t = 0$, we have that $H(x_1, \dots, x_n, 0) = f_1 \cdot f_2$. When $t = 1$, we have that $H(x_1, \dots, x_n, 1) = g_1 \cdot g_2$. Now notice that by definition of H_1 and H_2 , if one of $x_1, \dots, x_n$ is equal to 0 or 1, then H_1 and H_2 is constant and maps to x_0 . This means that H also has such property and hence H is a homotopy $(I, \partial I^n)$ to (X, x_0) .

We now have an appropriate binary operation on $\pi_n(X, x_0)$. It is clearly associative since the composition of maps are associativity and one can re-parametrize homotopies with different traversal speeds. I claim that the constant map $e_{x_0} : (I, \partial I^n) \rightarrow (X, x_0)$ defined by $e_{x_0}(x) = x_0$ is the identity. Let $f : (I^n, \partial I^n) \rightarrow (X, x_0)$ be arbitrary. Define the homotopy from $e_{x_0} \cdot f$ to f by

$$H(x_1, \dots, x_n, t) = \begin{cases} e_{x_0}(x_1, \dots, x_n) = x_0 & \text{if } 0 \leq x_1 \leq \frac{1-t}{2} \\ f\left(\frac{2s+t-1}{t+1}\right) & \text{if } \frac{1-t}{2} \leq x_1 \leq 1 \end{cases}$$

A similar homotopy proves that $f \cdot e_{x_0} \simeq f$. For the inverse, I claim that $\bar{f} : (I^n, \partial I^n) \rightarrow (X, x_0)$ defined by $\bar{f}(1 - x_1, \dots, x_n)$ is the inverse of f . Indeed, define a homotopy from $f \cdot \bar{f}$ to e_{x_0} by

$$H(x_1, \dots, x_n, t) = \begin{cases} e_{x_0}(x_1, \dots, x_n) = x_0 & \text{if } 0 \leq x_1 \leq \frac{t}{2} \text{ or } \frac{1-t}{2} \leq x_1 \leq 1 \\ f(2x_1 - t, x_2, \dots, x_n) & \text{if } \frac{t}{2} \leq x_1 \leq \frac{1}{2} \\ \bar{f}(2s + t - 1) & \text{if } \frac{1}{2} \leq x_1 \leq \frac{1-t}{2} \end{cases}$$

However, what makes each $\pi_n(X, x_0)$ for $n \geq 2$ different from the fundamental group $\pi_1(X, x_0)$ is the abelian group structure on $\pi_n(X, x_0)$.

Theorem 1.1.6 Let (X, x_0) be a pointed space. Then the n th homotopy group

$$\pi_n(X, x_0)$$

together with concatenation is abelian.

Proof Define a new operation $\star : \pi_n(X, x_0) \times \pi_n(X, x_0) \rightarrow \pi_n(X, x_0)$ by

$$[f] \star [g] = \begin{cases} f(t_1, 2t_2, \dots, t_n) & \text{if } 0 \leq t_1 \leq \frac{1}{2} \\ g(t_1, 2t_2 - 1, \dots, t_n) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

Such an operation clearly also defines an abelian group structure on $\pi_n(X, x_0)$ using the same argument. Now I want to prove that

$$([f] \star [g]) \star ([h] \star [k]) = ([f] \star [h]) \star ([g] \star [k])$$

This is true because

$$([f] \star [g]) \star ([h] \star [k]) = \begin{cases} f(2x_1, 2x_2, x_3, \dots, x_n) & \text{if } 0 \leq x_1, x_2 \leq \frac{1}{2} \\ g(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \text{ and } \frac{1}{2} \leq x_2 \leq 1 \\ h(2x_1 - 1, 2x_2, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq \frac{1}{2} \\ k(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1, x_2 \leq 1 \end{cases}$$

and

$$([f] \star [h]) \star ([g] \star [k]) = \begin{cases} f(2x_1, 2x_2, x_3, \dots, x_n) & \text{if } 0 \leq x_1, x_2 \leq \frac{1}{2} \\ h(2x_1 - 1, 2x_2, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq \frac{1}{2} \\ g(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } 0 \leq x_1 \leq \frac{1}{2} \text{ and } \frac{1}{2} \leq x_2 \leq 1 \\ k(2x_1, 2x_2 - 1, x_3, \dots, x_n) & \text{if } \frac{1}{2} \leq x_1, x_2 \leq 1 \end{cases}$$

which are entirely the same. Now I claim that $\star = \star$. It is clear that both binary operations have the same identity element e_{x_0} . Now we have that

$$f \star g = (f \star 1) \star (1 \star g) = (f \star 1) \star (1 \star g) = f \star g$$

Finally, I claim that $\star$ is commutative. We have that

$$f \star g = (1 \star f) \star (g \star 1) = (1 \star g) \star (f \star 1) = g \star f = g \star f$$

Thus we conclude. ■

The above technique is actually called the Eckmann-Hilton argument. In particular, it shows that concatenation of paths need not be defined via the first coordinate. Any choice of coordinate to perform concatenation will result in the same group structure.

Example 1.1.7

The homotopy groups of the sphere is given by

$$\pi_n(S^1) \cong \begin{cases} \mathbb{Z} & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof

We have seen that $\pi_0(S^1) = 0$ and $\pi_1(S^1) \cong \mathbb{Z}$.

When $n \geq 2$, suppose that $[f] \in \pi_n(S^1)$. Then we have a map $f : S^k \rightarrow S^1$. By the lifting criterion in Algebraic Topology 1, we may lift the map to the covering space $p : \mathbb{R} \rightarrow S^1$ since $0 = f_*(\pi_1(S^k)) \leq p_*(\pi_1(\mathbb{R})) = 0$. Hence f factors through a contractible space, and so is null homotopic. Hence $[f] = 0$. ■

Lemma 1.1.8

For any $n \in \mathbb{N}^+$, the two spaces $(I^n, \partial I^n)$ and (S^n, s_0) are homotopy equivalent.

Corollary 1.1.9

Let (X, x_0) be a pointed space. Then we have

$$\pi_n(X, x_0) = [(S^n, 1), (X, x_0)]_*$$

Therefore an alternate viewpoint of the homotopy groups is instead the collection of maps from the pointed n -sphere to the space X quotient homotopy. Indeed an n -dimensional sphere has an n -dimensional hole enclosed by the sphere itself. Therefore in order to detect n -dimensional holes in a space, we are permitted to try and fit n -spheres into the space.

Spheres are also advantageous for the definition of π_n because spheres only has an n -dimensional hole and no other holes in any dimension. Therefore we are capturing the minimal amount of information on n -dimensional holes without producing excess data.

The group structure of π_n is also easy to understand in terms of spheres.

Lemma 1.1.10

Let (X, x_0) be a pointed space. Let $f, g : S^n \rightarrow X$ be pointed maps. Then $[f] \cdot [g] \in \pi_n(X, x_0)$ is represented by the map

$$S^n \xrightarrow{\text{proj.}} S^n \wedge S^n \xrightarrow{f \wedge g} X$$

where the projection map is given by $S^n \wedge S^n \cong \frac{S^{2n}}{S^{2n-1}}$ where $S^{n-1} \hookrightarrow S^n$ is the inclusion along the equator.

1.2 Functorial Properties of Homotopy

The homotopy groups also satisfy functorial properties similar to the fundamental group and the (co)homology groups.

Theorem 1.2.1 (Functoriality) Let (X, x_0) and (Y, y_0) be pointed spaces and let $f : (X, x_0) \rightarrow (Y, y_0)$ be a pointed map. Then the induced map

$$\pi_n(f) : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

defined by $[\gamma] \mapsto [f \circ \gamma]$ is a group homomorphism. Moreover, it satisfies the following functorial properties.

- If $g : (Y, y_0) \rightarrow (Z, z_0)$ is a pointed map then

$$\pi_n(g \circ f) = \pi_n(g) \circ \pi_n(f)$$

- If $\text{id}_{(X, x_0)} : (X, x_0) \rightarrow (X, x_0)$ is the identity map then

$$\pi_n(\text{id}_{(X, x_0)}) = \text{id}_{\pi_n(X, x_0)}$$

Proof Firstly, let us show that it is a group homomorphism. Let $\gamma_1, \gamma_2 \in \pi_n(X, x_0)$. We have that

$$\pi_n(f)([\gamma_1] \cdot [\gamma_2]) = [f \circ (\gamma_1 \cdot \gamma_2)] = [f \circ \gamma_1 \cdot f \circ \gamma_2] = \pi_n(f)([\gamma_1]) \cdot \pi_n(f)([\gamma_2])$$

where the second equality is true because homotopies are preserved under function composition. It remains to show associativity and unitality.

- Associativity: We have that

$$\pi_n(g \circ f)([\gamma]) = [g \circ f \circ \gamma] = \pi_n(g)([f \circ \gamma]) = (\pi_n(g) \circ \pi_n(f))([\gamma])$$

- Unitality: We have that

$$\pi_n(\text{id}_{(X, x_0)})([\gamma]) = [\text{id}_{(X, x_0)} \circ \gamma] = [\gamma] = \text{id}_{\pi_n(X, x_0)}([\gamma])$$

And so we conclude. ■

Similar to all other functorial properties we have seen throughout algebraic topology, a homeomorphism of spaces give an isomorphism on homotopy groups. Now that we know about category theory, we see that such a result does not depend on the definition of the homotopy groups or the (co)homology groups, but is in fact due to the functorial properties of each invariant.

Similar to (co)homology and the fundamental group, the homotopy groups are defined via a quotient with homotopy. Therefore we expect the homotopy groups to not be able to distinguish between homotopy equivalent spaces but not homeomorphic spaces.

Theorem 1.2.2 (Homotopy Equivalence) Let $(X, x_0), (Y, y_0)$ be pointed spaces and $f, g : (X, x_0) \rightarrow (Y, y_0)$ be pointed maps. If f and g are homotopic, then the induced maps

$$\pi_n(f) = \pi_n(g) : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

are equal. Moreover, if f is a homotopy equivalence, then $\pi_n(f)$ is an isomorphism.

Proof Let $[\gamma] \in \pi_n(X, x_0)$. Suppose that f and g are homotopic via $F : X \times I \rightarrow Y$. now define

$$H(x_1, \dots, x_n, t) = F(\gamma(x_1, \dots, x_n), t)$$

Then it is clear that $H(x_1, \dots, x_n, 0) = f \circ \gamma$ and $H(x_1, \dots, x_n, 1) = g \circ \gamma$. Thus $[f \circ \gamma] = [g \circ \gamma]$ and so we conclude that $\pi_n(f)([\gamma]) = \pi_n(g)([\gamma])$.

If f is a homotopy equivalence, then there exists $g : (Y, y_0) \rightarrow (X, x_0)$ such that $g \circ f \simeq \text{id}_{(X, x_0)}$ and $f \circ g \simeq \text{id}_{(Y, y_0)}$. By functoriality and homotopy equivalence, we have that

$$\pi_n(g) \circ \pi_n(f) = \text{id}_{\pi_n(X, x_0)} \quad \text{and} \quad \pi_n(f) \circ \pi_n(g) = \text{id}_{\pi_n(Y, y_0)}$$

and so we conclude. ■

Similar to the fundamental group, changing the base point via a path induces isomorphisms on homotopy groups with the same space but different base point.

Theorem 1.2.3 Let (X, x_0) and (X, x_1) be pointed spaces with the same base space. Let $u : I \rightarrow X$ be a path from x_0 to x_1 . Define the induced map

$$u_{\#} : \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$$

as follows. For $[\gamma] \in \pi_n(X, x_0)$ define $u_{\#}([\gamma])$ by first shrinking the domain of γ to a smaller concentric cube in I^n . Then inserting the path γ on each radical segment of the shell between the smaller cube and ∂I^n .

The construction of $u_{\#}$ is a group isomorphism. Moreover, it satisfies the following universal properties.

- If $v : I \rightarrow X$ is a path from x_1 to x_2 and $u \cdot v$ is the concatenation of these paths, then

$$(u \cdot v)_{\#} = u_{\#} \circ v_{\#}$$

- If c_{x_0} is the constant path from x_0 to x_0 then $(c_{x_0})_{\#}$ is the identity

Proposition 1.2.4 Let X_i for $i \in I$ be a family of path connected spaces. Then there are isomorphisms

$$\pi_n \left(\prod_{i \in I} X_i \right) \cong \prod_{i \in I} \pi_n(X_i)$$

Proposition 1.2.5 Let (X, x_0) and (X, x_1) be pointed spaces with the same base space. Let $u, v : I \rightarrow X$ be paths from x_0 to x_1 . If u and v are homotopic relative to end points then the induced

maps

$$u_{\#} = v_{\#} : \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$$

are equal.

This shows that if X is path connected, then $\pi_n(X, x_0)$ no longer depends on the choice of base point. Although there are no canonical isomorphisms between $\pi_n(X, x_0)$ and $\pi_n(X, x_1)$, we still forget about the base point in this case and write the homotopy groups as $\pi_n(X)$.

1.3 The Action of the Fundamental Group

Proposition 1.3.1

Let (X, x_0) be a pointed space. Let $n \geq 1$. Then the map $\pi_1(X, x_0) \times \pi_n(X, x_0) \rightarrow \pi_n(X, x_0)$ defined by

$$([u], [f]) \mapsto u_{\#}(f)$$

is a group action on $\pi_n(X, x_0)$.

When $n = 1$, the above group action is just left multiplication in $\pi_1(X, x_0)$.

1.4 Relative Homotopy Groups

Definition 1.4.1 (Triplets of Spaces) Let X be a topological space. A pointed pair of space is a triple (X, A_1, A_2) where $A_2 \subseteq A_1 \subseteq X$ are subspaces of X . A map between triplets of spaces $f : (X, A_1, A_2) \rightarrow (Y, B_1, B_2)$ is a map $f : X \rightarrow Y$ such that $f(A_1) \subseteq B_1$ and $f(A_2) \subseteq B_2$. If $A_2 = \{x_0\}$ is a single point we say that (X, A, x_0) is a pointed pair of spaces.

Definition 1.4.2 (Homotopy between Maps of Triplets) Let $f, g : (X, A_1, A_2) \rightarrow (Y, B_1, B_2)$ be maps triplets of spaces. A homotopy between f and g is a homotopy between $f : X \rightarrow Y$ and $g : X \rightarrow Y$, namely $H : X \times [0, 1] \rightarrow Y$ such that $H(A_1 \times [0, 1]) \subseteq B_1$ and $H(A_2 \times [0, 1]) \subseteq B_2$.

Definition 1.4.3 (The n th Relative Homotopy Groups) Let (X, A, x_0) be a pointed pair of space. Let $n \geq 2$. Regard I^{n-1} sitting inside I^n by $I^{n-1} = \{(x_1, \dots, x_n) \in I^n \mid x_n = 0\}$ and let $J^{n-1} = \partial I^n \setminus I^{n-1}$. Define the relative homotopy groups of the triple by

$$\pi_n(X, A, x_0) = \frac{\left\{ \gamma : (I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0) \mid \gamma \text{ is continuous} \right\}}{\simeq}$$

where we say that $f \simeq g$ if there exists a homotopy between f and g .

It is easy to see that $\pi_n(X, x_0, x_0) = \pi_n(X, x_0)$ so that homotopy groups are a special case of the relative homotopy groups.

Lemma 1.4.4 For any $n \in \mathbb{N}$, the two triplets $(I^n, \partial I^n, J^{n-1})$ and (D^n, S^{n-1}, s_0) are homotopy equivalent.

Theorem 1.4.5 Let (X, A, x_0) be a pointed pair of space. The composition law on definition 1.1.4 defines a group structure on $\pi_n(X, A, x_0)$ for $n \geq 2$. Moreover, $\pi_n(X, A, x_0)$ is abelian for $n \geq 3$.

1.5 Induced Maps of Relative Homotopy Groups

Theorem 1.5.1 Let (X, A, x_0) and (Y, B, y_0) be pointed pairs of spaces and $f : (X, A, x_0) \rightarrow (Y, B, y_0)$ a map. Then f induces a map on the relative homotopy groups

$$f_* : \pi_n(X, A, x_0) \rightarrow \pi_n(Y, B, y_0)$$

for $n \geq 2$ satisfying the following functorial properties:

- f_* is a group homomorphism
- If $g : (Y, B, y_0) \rightarrow (Z, C, z_0)$ is a map, then

$$(g \circ f)_* = g_* \circ f_*$$

- If $\text{id}_{(X, A, x_0)}$ is the identity map on (X, A, x_0) , then

$$(\text{id}_{(X, A, x_0)})_* = \text{id}_{\pi_n(X, A, x_0)}$$

Theorem 1.5.2 Let $(X, A, x_0), (Y, B, y_0)$ be pointed pairs of spaces and $f, g : (X, A, x_0) \rightarrow (Y, B, y_0)$ be pointed maps. If f and g are homotopic, then the induced maps

$$f_* = g_* : \pi_n(X, A, x_0) \rightarrow \pi_n(Y, B, y_0)$$

are equal. Moreover, if f is a homotopy equivalence, then f_* is an isomorphism.

TBA: change of base point isomorphisms.

Theorem 1.5.3 (The Hurewicz Homomorphism) Let (X, A, x_0) be a pointed pair of space. Let u_n be a generator of $H_n(S^n) \cong \mathbb{Z}$. Then the map

$$h : \pi_n(X, A, x_0) \rightarrow H_n(X, A)$$

defined by $[f] \mapsto f_*(u_n)$ is a group homomorphism.

1.6 Long Exact Sequence in Homotopy Groups

Lemma 1.6.1 (Compression Criterion) Let (X, A, x_0) be a pair of spaces with basepoint. Let $f : (D^n, S^{n-1}, *) \rightarrow (X, A, x_0)$ be a map. Then $[f] = [e_{x_0}] \in \pi_n(X, A, x_0)$ if and only if

$$(f : D^n \rightarrow X) \stackrel{S^{n-1}}{\simeq} (g : D^n \rightarrow X)$$

where g is any map such that $g(X) \subseteq A$.

Proof Suppose that the second criterion is satisfied. Then it clearly shows that $[f] = [g] \in \pi_n(X, A, x_0)$. Let $r : D^n \times I \rightarrow D^n$ be a deformation retract from D^n to $*$ in $S^{n-1} \subset D^n$. Consider the map $g \circ r : D^n \times I \rightarrow X$. When $t = 0$, this is the map g . When $t = 1$, $g \circ r$ factors through $*$ and so becomes a map $*$ $\rightarrow$ X . In other words, it is the constant map e_{x_0} . Moreover, it $g \circ r$ has image in A and so in particular it sends S^{n-1} to A . Thus $g \circ r$ is a homotopy between e_{x_0} and g . We conclude that $[f] = [g] = [e_{x_0}]$.

Now suppose that $[f] = [e_{x_0}] \in \pi_n(X, A, x_0)$ is given by the homotopy $H : D^n \times I \rightarrow X$. This means that $H(D^n \times \{1\}) \subseteq \{x_0\} \subset A$ and $H(S^{n-1} \times I) \subset A$. Now $D^n \times I$ deformation retracts to the cup $D^n \times \{1\} \cup S^{n-1} \times I$ by radial projection from the center point of $D^n \times \{0\}$. Thus H can be converted into a map from $D^n \times \{1\} \cup S^{n-1} \times I$ to X . Then H is now a homotopy from f to a map $H(-, 1) : D^n \rightarrow X$ which has image in A , relative to S^{n-1} . Thus we conclude. ■

Theorem 1.6.2 Let X be a space and A, B be subspaces of X such that $B \subseteq A \subseteq X$. Let $x_0 \in B$. Then there is a long exact sequence in relative homotopy groups:

$$\cdots \longrightarrow \pi_n(A, B, x_0) \xrightarrow{i_*} \pi_n(X, B, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial_n} \pi_{n-1}(A, B, x_0) \longrightarrow \cdots \longrightarrow \pi_1(X, A, x_0)$$

where $i : (A, B, x_0) \rightarrow (X, B, x_0)$ and $j : (X, B, x_0) \rightarrow (X, A, x_0)$ are the inclusions and $\partial : \pi_n(X, A, x_0) \rightarrow \pi_{n-1}(A, B, x_0)$ is given by $[\gamma] \mapsto [\gamma|_{I^{n-1}}]$

Proof ■

TBA: Naturality of the sequence.

Theorem 1.6.3 Let (X, A, x_0) be a pointed pair of spaces. The relative homotopy groups and (absolute) homotopy groups of (X, A, x_0) fit into a long exact sequence

$$\cdots \longrightarrow \pi_{n+1}(X, A, x_0) \xrightarrow{\partial_{n+1}} \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial_n} \pi_{n-1}(A, x_0) \longrightarrow \cdots \longrightarrow \pi_0(X, x_0) \longrightarrow 0$$

where ∂_n is defined by $[f] \mapsto [f|_{I^{n-1}}]$ and i_* and j_* are induced by inclusions.

Note that even though at the end of the sequence group structures are not defined, exactness still makes sense: kernels in this case consists of elements that map to the homotopy class of the constant map.

Theorem 1.6.4 Let $p : E \rightarrow B$ be a fiber bundle. Let $A \subseteq B$. Let $y_0 \in E$ and $p(y_0) = x_0$. Then there is an isomorphism

$$\pi_n(E, p^{-1}(A), y_0) \cong \pi_n(B, A, x_0)$$

given by the induced map p_* for all $n \geq 2$.

1.7 Homotopical Connectivity

Definition 1.7.1 (n-Connected Space) Let X be a space. We say that it is n -connected if

$$\pi_k(X, x_0) = 0$$

for $0 \leq k \leq n$ and some $x_0 \in X$.

Note that $\pi_0(X, x_0)$ implies that X is path connected. Hence the notion of n -connectedness does not depend on the base point by the change of base point isomorphism. In particular, $\pi_k(X, x_0) = 0$ for $0 \leq k \leq n$ and some $x_0 \in X$ if and only if $\pi_k(X, x_0) = 0$ for $0 \leq k \leq n$ for all $x_0 \in X$. (Hatcher)

Definition 1.7.2 (n-Connected Pair of Spaces) Let (X, A) be a pair of space. We say that it is n -connected if the following are true.

- $\pi_k(X, A, x_0) = 0$ for $0 < k \leq n$ and all $x_0 \in A$.
- $\pi_0(\iota) : \pi_0(A) \rightarrow \pi_0(X)$ is surjective.

TBA: conditions in P.346 of Hatcher

Definition 1.7.3 (Weakly Contractible) Let X be a space. We say that X is weakly contractible if

$$\pi_n(X) = 0$$

for all $n \geq 0$.

2 Weak Equivalences and CW-Complexes

2.1 Weak Homotopy Equivalence

Definition 2.1.1 (Weak Homotopy Equivalence) We say that a map $f : X \rightarrow Y$ is a weak homotopy equivalence if it induces isomorphisms on all homotopy groups π_n on any choice of base point.

TBA: compression lemma in Hatcher

Theorem 2.1.2 Let X, Y be spaces and let $f : X \rightarrow Y$ be a weak homotopy equivalence. Then f induces isomorphisms

$$f_* : H_n(X; G) \xrightarrow{\cong} H_n(Y; G) \quad \text{and} \quad f^* : H^n(Y; G) \xrightarrow{\cong} H^n(X; G)$$

for any group G and all $n \in \mathbb{N}$.

This theorem shows that the higher homotopy groups is not a weaker invariant than homology and cohomology. Indeed, the theorem states that if the all homotopy groups are isomorphic, then all their (co)homology groups will be isomorphic.

Proposition 2.1.3 Let X, Y be spaces and let $f : X \rightarrow Y$ be a weak homotopy equivalence. Then f induces bijections

$$[Z, X] \cong [Z, Y] \quad \text{and} \quad [Z, X]_* \cong [Z, Y]_*$$

for all CW-complexes Z .

2.2 Whitehead's Theorem

Theorem 2.2.1 (Whitehead's Theorem) If X and Y are CW-complexes and $f : X \rightarrow Y$ is a weak homotopy equivalence, then f is a homotopy equivalence.

TBA: extension lemma in Hatcher.

Corollary 2.2.2 If X and Y are CW-complexes with $\pi_1(X) = \pi_1(Y) = 0$ and $f : X \rightarrow Y$ induces isomorphisms on homology groups H_n for all n , then f is a homotopy equivalence.

2.3 Cellular Approximations

Definition 2.3.1 (Cellular Maps)

Let X and Y be CW-complexes. A map $f : X \rightarrow Y$ is called cellular if $f(X_n) \subset Y_n$ for all n , where X_n is the n -skeleton of X .

Definition 2.3.2 (Cellular Approximations) Let X and Y be CW-complexes. We say that $f : X \rightarrow Y$ has a cellular approximations if f is homotopic to a cellular map $f' : X \rightarrow Y$.

To this end we need to revisit the notion of polyhedra.

Definition 2.3.3 (Convex Polyhedra) Let $n \in \mathbb{N}$. A convex polyhedra is a subset S of $\mathbb{R}^n$ of the form

$$S = \left\{ x = (x_1, \dots, x_n) \in \mathbb{R}^n \mid \sum_{k=1}^n a_{k,1}x_k \leq b_1, \dots, \sum_{k=1}^n a_{k,s}x_k \leq b_s \right\}$$

for some $a_{1,1}, \dots, a_{n,s}, b_1, \dots, b_s \in \mathbb{R}$.

Lemma 2.3.4 Let X be a CW complex. Let $Y \subseteq X$ be a CW subcomplex. Let Z be obtained by attaching a cell e^k to Y . Let $f : I^n \rightarrow Z$ be a map. Then there exists a map $g : I^n \rightarrow X$ such that

$$f \stackrel{f^{-1}(Y)}{\simeq} g$$

and g is such that the following is true. There exists a simplex $\Delta^k \subset e^k$ and polyhedra $P_1, \dots, P_d$ such that g is the (possibly empty) union

$$g^{-1}(\Delta^k) = \bigcup_{k=1}^n$$

and $g|_{P_t}$ is the restriction of a linear surjection $\mathbb{R}^n \rightarrow \mathbb{R}^k$ for all t .

Theorem 2.3.5 (Cellular Approximation Theorem)

Let X, Y be CW complexes. Let $f : X \rightarrow Y$ be a map. Then the following are true.

- f is homotopic to a cellular map (called its cellular approximation).
- If f is homotopic to a cellular map f' and f is already cellular for some subcomplex $A \subseteq X$, then the homotopy to its cellular approximation can be taken such that $f'|_A = f|_A$.

Example 2.3.6

Let $0 \leq n < m$. Then $\pi_n(S^m) = 0$.

Proof

Let $[f] \in \pi_n(S^m)$. By the above theorem, we know that $f : S^n \rightarrow S^m$ is homotopic to a cellular map, where S^n is given the CW structure of a 0-cell and an n -cell. But the n -skeleton of S^m only consists of its 0-cell. Hence the cellular approximation of f is null-homotopic. Hence $[f] = 0$. ■

Example 2.3.7

$S^\infty = \text{colim}_{n \in \mathbb{N}} S^n$ is weakly contractible and contractible.

Proof

Let $n \in \mathbb{N}$. Let $[f] \in \pi_n(S^\infty)$. Then $f : S^n \rightarrow S^\infty$ has a compact image since S^n is compact. By definition of the colimit, we have that $\{S^n \mid n \in \mathbb{N}\}$ is an open cover of S^∞ , hence the image of f lies in $S^{n_1}, \dots, S^{n_k}$ for some $n_1, \dots, n_k \in \mathbb{N}$. Let $m = \max\{n_1, \dots, n_k, n + 1\}$. Since the colimit is a direct limit and each map is an inclusion, we have that f factors into the following:

$$\begin{array}{ccc} S^n & \xrightarrow{f} & S^\infty \\ & \searrow g & \nearrow \\ & S^m & \end{array}$$

But $m > n$ since $m \geq n + 1$, and we know that $\pi_n(S^m) = 0$ for $m > n$. Hence g is null homotopic, and so is f . Since S^∞ is a CW complex, by Whitehead's theorem, S^∞ is contractible. ■

Theorem 2.3.8 (Relative Cellular Approximation) Any map $f : (X, A) \rightarrow (Y, B)$ between pairs of CW-complexes has a cellular approximation.

Corollary 2.3.9 Let $A \subset X$ be CW-complexes and suppose that all cells $X \setminus A$ have dimension larger than n . Then (X, A) is n -connected.

Corollary 2.3.10 Let X be a CW complex and let X^n be its n -skeleton. Then (X, X^n) is n -connected. Moreover, the inclusion $X^n \hookrightarrow X$ induces an isomorphism

$$\pi_k(X^n) \rightarrow \pi_k(X)$$

for $0 \leq k < n$ and a surjection for $k = n$.

2.4 CW Approximations

Definition 2.4.1 (CW Approximation) Let X be a space. A CW approximation of X is a weak homotopy equivalence $f : Z \rightarrow X$ where Z is a CW complex.

The goal of this section is that every space has a CW approximation. The given homotopy equivalence makes this notion powerful because this means that for any space X , there exists a CW-complex such that X and Z are homotopy equivalent, and moreover, has isomorphic homotopy, homology and cohomology groups.

Definition 2.4.2 (CW Model) Let (X, A) be a non-empty pair of CW-complexes. An n -connected CW model of (X, A) is an n -connected CW pair (Z, A) together with a map $f : Z \rightarrow X$ with $f|_A = \text{id}_A$ such that

$$f_* : \pi_i(Z) \rightarrow \pi_i(X)$$

is an isomorphism for $i > n$ and an injection for $i = n$ for any choice of base point.

Theorem 2.4.3 For any non-empty pair (X, A) of CW-complexes, there exists an n -connected model (Z, A) . Moreover, Z can be built from A by attaching cells of dimension greater than n .

Theorem 2.4.4 Every pair of spaces (X, A) has a CW approximation. Such a CW approximation is unique up to homotopy equivalence.

3 Main Results of Homotopy Theory on CW-Complexes

3.1 Excision for Homotopy Groups

Theorem 3.1.1 (The Homotopy Excision Theorem (Blaker's Massey Theorem)) Let X be a CW-complex and A, B be sub complexes such that $X = A \cup B$ and $A \cap B \neq \emptyset$. If $(A, A \cap B)$ is m -connected and $(B, A \cap B)$ is n -connected for $m, n \geq 0$, then the map

$$\pi_k(\iota) : \pi_k(A, A \cap B) \rightarrow \pi_k(X, B)$$

induced by the inclusion $\iota : (A, A \cap B) \rightarrow (X, B)$ is an isomorphism for $0 \leq k < m + n$ and a surjection for $k = m + n$.

Proof We prove this by considering successively more general cases, starting from the simplest one.

Case 1: A is obtained from $A \cap B$ by attaching some e_α^{m+1} -cells and B is obtained from $A \cap B$ by attaching one e^{n+1} -cell.

We want to show that the map $\pi_k(A, A \cap B) \rightarrow \pi_k(X, B)$ is surjective. Let $f : (I^k, \partial I^k, J^{k-1}) \rightarrow (X, B, x_0)$ represent an equivalence class in $\pi_k(X, B)$. Since f is continuous, f preserves compactness and hence the image of f is compact. By a property of CW complexes, $\text{im}(f)$ meets only finitely many of the cells e_α^{m+1} and e^{n+1} . ■

Proposition 3.1.2 Let (X, A) be a pair of r -connected CW complexes and let A be s -connected. Then the map

$$p_* : \pi_k(X, A) \rightarrow \pi_k(X/A)$$

induced by the quotient map $p : X \rightarrow X/A$ is an isomorphism for $0 \leq k \leq r + s$ and a surjection for $k = r + s + 1$.

3.2 Hurewicz's Theorem

Theorem 3.2.1 (Hurewicz's Homomorphism) Let X be a path connected space. Then for any $n \in \mathbb{N}$, there is a group homomorphism

$$h_n : \pi_n(X) \rightarrow H_n(X)$$

called the Hurewicz homomorphism, defined as follows. Let $[u_n] \in H_n(S^n)$ be a canonical generator. Then $h_n([f]) = f_*(u_n)$.

Theorem 3.2.2 (Hurewicz's Theorem) Let X be a space. Then the following are true regarding Hurewicz's homomorphism.

- Let $n \geq 2$. If X is $(n - 1)$ -connected, then $\tilde{H}_k(X) = 0$ for all $0 \leq k < n$. Moreover, the Hurewicz homomorphism

$$h_n : \pi_n(X) \rightarrow H_n(X)$$

is an isomorphism. Moreover, h_{n+1} is a surjection.

- Let $n = 1$, then Hurewicz's homomorphism induces an isomorphism

$$\overline{h}_1 : \pi_1(X)^{\text{ab}} \rightarrow H_1(X)$$

Theorem 3.2.3 (Relative Hurewicz's Homomorphism) Let (X, A) be a pair of spaces. Then for any $n \geq 1$, there is a group homomorphism

$$h_n : \pi_n(X, A) \rightarrow H_n(X, A)$$

called the relative Hurewicz homomorphism, defined as follows. Let $[u_n] \in H_n(S^n, \partial S^n)$ be a canonical generator. Then $h_n([f]) = f_*(u_n)$.

Theorem 3.2.4 (Relative Hurewicz's Theorem) Let (X, A) be a pair of spaces. Let $n \geq 2$. If X and A are path connected and (X, A) is $(n - 1)$ -connected, then $H_k(X, A) = 0$ for all $0 \leq k < n$. Moreover, the Hurewicz homomorphism

$$h_n : \pi_n(X, A, x_0) \rightarrow H_n(X, A)$$

is an isomorphism.

Theorem 3.2.5 (Naturality of Hurewicz's Homomorphism) Let (X, x_0) and (Y, y_0) be pointed spaces and let $f : (X, x_0) \rightarrow (Y, y_0)$ be a map. Then the following diagram is commutative:

$$\begin{array}{ccc} \pi_k(X, x_0) & \xrightarrow{\pi_k(f)} & \pi_k(Y, y_0) \\ h_k \downarrow & & \downarrow h_k \\ H_k(X) & \xrightarrow{f_*} & H_k(Y) \end{array}$$

where h is the Hurewicz homomorphism. Moreover, a similar diagram is also commutative for the relative Hurewicz homomorphism.

The connection between the homotopy groups and the homology groups begs the question of whether there is a relationship between the homotopy groups and cohomology groups that is not implicit by the relation between homology and cohomology. This is answered in Stable Homotopy Theory, when we introduced Brown's representability theorem.

3.3 Eilenberg-MacLane Spaces

Definition 3.3.1 (Eilenberg-MacLane Space) Let G be a group and $n \in \mathbb{N}$. We say that a space X is an Eilenberg-MacLane space of type $K(G, n)$ if

$$\pi_k(X) = \begin{cases} K(G, n) & \text{if } k = n \\ 0 & \text{otherwise} \end{cases}$$

We often denote this space X directly by $X = K(G, n)$.

Proposition 3.3.2 Let G be a group. Then there exists a $K(G, 1)$ -CW complex.

Theorem 3.3.3 Let G be an abelian group and $n \geq 2$. Then there exists a $K(G, n)$ -CW complex. Moreover, it is uniquely determined by G and n .

The Eilenberg-MacLane spaces are a fundamental object of study in algebraic topology because it is a universal object. This is again part of Stable Homotopy Theory and is the same theorem that gives the connection between homotopy groups and cohomology groups.

Theorem 3.3.4 (Brown's Representability Theorem for Singular Cohomology) Let G be an abelian group. Then there is a natural isomorphism

$$H^n(-; G) \cong [-, K(G, n)]$$

for each $n \in \mathbb{N}$.

4 Generalized (Co)Homology Theories

4.1 Weak Equivalences and (Co)Homological Isomorphisms

Proposition 4.1.1 Let X, Y be spaces. Let A be an abelian group. Let $f : X \rightarrow Y$ be a weak equivalence. Then

$$f_* : H_n(X; A) \rightarrow H_n(Y; A) \quad \text{and} \quad f^* : H^n(Y; A) \rightarrow H^n(X; A)$$

are isomorphisms for all n .

4.2 Generalized Homology Theories

Definition 4.2.1 (Generalized Homology Theory for Spaces) A Generalized Homology Theory is a collection of functors

$$h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab} \quad \text{and} \quad \delta_n : h_n \circ F \Rightarrow h_{n-1}$$

where $F(X, Y) = (Y, \emptyset)$, for each $n \in \mathbb{N}$, satisfying the first four axioms together with the following.

- Weak Equivalence: If $f : (X, A) \rightarrow (Y, B)$ is a weak equivalence, then

$$f_* : h_n(X, A) \rightarrow h_n(Y, B)$$

is an isomorphism.

By adding on the axiom of weak equivalence and the fact that every space admits a weak equivalence to a CW complex, we can see that the two theories are the same.

Theorem 4.2.2 Any generalized homology theory on $\mathbf{Top}^2$ determines and is determined by a generalized homology theory on $\mathbf{CW}^2$.

Definition 4.2.3 (Ordinary Homology Theory) Let G be an abelian group. If a generalized homology theory (h_n, δ_n) in addition satisfies

- Dimension:

$$h_n(*) = \begin{cases} G & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Then h_n is called an ordinary homology theory.

Theorem 4.2.4 (Eilenberg-Steenrod Uniqueness Theorem) Let $T : (h_n, \delta_n) \rightarrow (h'_n, \delta'_n)$ be a natural transformation of generalized homology theories defined on $\mathbf{CW}^2$ such that $h_n(*) \cong h'_n(*)$, then T is a natural isomorphism

$$(h_n, \delta_n) \cong (h'_n, \delta'_n)$$

4.3 Generalized Cohomology Theories

Definition 4.3.1 (Generalized Cohomology Theory for Spaces) A Generalized cohomology theory is a collection of contravariant functors and natural transformations

$$h^n : \mathbf{Top}^2 \rightarrow \mathbf{Ab} \quad \text{and} \quad \delta^n : h^n \circ F \Rightarrow h^{n+1}(X, A)$$

where $F(X, A) = (A, \emptyset)$, for each $n \in \mathbb{N}$, satisfying the above first four axioms and the following.

- Weak Equivalence: If $f : (X, A) \rightarrow (Y, B)$ is a weak equivalence, then

$$f_* : h^n(Y, B) \rightarrow h^n(X, A)$$

is an isomorphism.

Proposition 4.3.2 Let F and G be two generalized cohomology theories for CW pairs. Let $\lambda : F \Rightarrow G$ be a natural transformation such that

$$\lambda : F(*, \emptyset) \xrightarrow{\cong} G(*, \emptyset)$$

is an isomorphism for $*$ the one-point space. Then λ is a natural isomorphism.

5 The Stable Phenomena

5.1 Freudenthal Suspension Theorem

Theorem 5.1.1 (Freudenthal Suspension Theorem) Let X be an n -connected CW complex. Then for $0 \leq k \leq 2n$, the induced map

$$\Sigma_* : \pi_k(X) \rightarrow \pi_{k+1}(\Sigma X)$$

is an isomorphism. For $k = 2n + 1$, Σ_* is a surjection.

We can keep on suspending the space and the maps. Indeed if X is n -connected then, by Freudenthal suspension theorem ΣX is $(n + 1)$ -connected. We can then apply the suspension theorem again on ΣX and we see that $\Sigma^2 X$ is $(n + 2)$ -connected.

The following theorem is also said to be the Freudenthal suspension theorem.

Theorem 5.1.2 Let Y be $(n - 1)$ -connected. Consider the reduced suspension functor $\Sigma : \mathbf{hTop}_* \rightarrow \mathbf{hTop}_*$. Then $\Sigma : [X, Y] \rightarrow [\Sigma X, \Sigma Y]$ is bijective if $\dim(X) < 2n - 1$. Moreover, it is a surjection if $\dim(X) = 2n - 1$.

Corollary 5.1.3 There is an isomorphism

$$\pi_{n+k}(S^n) \cong \pi_{n+k+1}(S^{n+1})$$

for all $n \geq k + 2$.

Proposition 5.1.4 Let X be a space. Let $k \in \mathbb{N}$. Then the the following sequence of suspensions

$$\pi_k(X) \rightarrow \pi_{k+1}(\Sigma X) \rightarrow \pi_{k+2}(\Sigma^2 X) \rightarrow \dots$$

are eventually isomorphisms.

Proof Let X be n -connected. There are two cases.

Let $k \leq 2n$. By Freudenthal suspension theorem, if $k \leq 2n$ then $\pi_k(X) \cong \pi_{k+1}(\Sigma X)$. Then ΣX is $(n + 1)$ -connected hence $\pi_{k+1}(\Sigma X) \cong \pi_{k+2}(\Sigma^2 X)$ is an isomorphism since $k + 1 \leq 2n + 2$. More generally, for $r \in \mathbb{N}$, $\Sigma^r X$ is $(r + n)$ -connected hence

$$\pi_{k+r}(\Sigma^r X) \cong \pi_{k+r+1}(\Sigma^{r+1} X)$$

is an isomorphism since $k + r \leq 2n + 2r$.

Now if $k > 2n$, then there exists $r \in \mathbb{N}$ such that $k + r \leq 2n + 2r$. Such an r is given by say $k - 2n$. Then by Freudenthal suspension theorem,

$$\pi_{k+r}(\Sigma^r X) \cong \pi_{k+r+1}(\Sigma^{r+1} X)$$

is an isomorphism. More generally, for $m \in \mathbb{N}$, $\Sigma^{r+m} X$ is $(r + m + n)$ -connected hence

$$\pi_{k+r+m}(\Sigma^{r+m} X) \cong \pi_{k+r+m+1}(\Sigma^{r+m+1} X)$$

is an isomorphism since $k + r + m \leq 2n + 2r + 2m$. ■

5.2 The Stable Homotopy Groups

In Algebraic Topology, we refer to a property or invariant being stable if applying the suspension functor to the space does not change the property (up to possibly a shift in index). The first such example is one we have already encountered.

Definition 5.2.1 (Stable Homotopy Groups) Let X be a space. Let $n \in \mathbb{N}$. Define the n th stable homotopy groups of X to be

$$\pi_n^s(X) = \operatorname{colim}_{k \rightarrow \infty} \pi_{n+k}(\Sigma^k X)$$

This is well defined because of the Freudenthal suspension theorem, which states that the groups in the direct limit eventually stabilize. Indeed, the same theorem shows the following.

Proposition 5.2.2 Let X be a space. Then there is an isomorphism

$$\pi_n^s(X) \cong \pi_{n+1}^s(\Sigma X)$$

induced by the suspension functor between stable homotopy groups for any $n \in \mathbb{N}$.

Proof This can be seen by simply unwinding the definitions. We have that

$$\begin{aligned} \pi_{n+1}^s(\Sigma X) &= \operatorname{colim}_{k \rightarrow \infty} \pi_{n+k+1}(\Sigma^{k+1} X) \\ &= \operatorname{colim}_{k \rightarrow \infty} \pi_{n+k}(\Sigma^k X) \\ &= \pi_n^s(X) \end{aligned}$$

and so we conclude. ■

New: Graded ring structure on π_*^s .

Recall that for $n \geq 2$, the set $[\Sigma^2 X, Z]$ for any spaces X and Z are abelian. Moreover, the suspension map $\Sigma : [\Sigma^2 X, Z] \rightarrow [\Sigma^3 X, \Sigma Z]$ is a group homomorphism. In particular, we can choose $Z = \Sigma^2 Y$. Now since the category **Ab** of abelian groups is cocomplete, the following inverse system

$$[\Sigma^2 X, \Sigma^2 Y] \xrightarrow{\Sigma} [\Sigma^3 X, \Sigma^3 Y] \xrightarrow{\Sigma} [\Sigma^4 X, \Sigma^4 Y] \longrightarrow \dots$$

has an inverse limit. This leads to the following definition.

Definition 5.2.3 (Set of Stable Homotopy Classes of Maps) Let X and Y be space. The set of stable homotopy classes of maps from X to Y is defined to be the abelian group

$$[X, Y]^s = \operatorname{colim}_{n \in \mathbb{N} \setminus \{0,1\}} [\Sigma^n X, \Sigma^n Y]$$

The following observation is crucial. When X and Y are pointed CW complexes, the abelian groups $[\Sigma^n X, \Sigma^n Y]$ also become stable under suspensions.

Theorem 5.2.4 Let X and Y be pointed CW complexes. Then the sequence of abelian groups,

$$[\Sigma^n X, \Sigma^n Y] \xrightarrow{\Sigma} [\Sigma^{n+1} X, \Sigma^{n+1} Y] \longrightarrow \dots$$

are isomorphic under the suspension functor for $n > \dim(X)$ and $n \geq 2$.

New: When X is compact, $[X, QY]_* = [X, Y]_*^s$.

Theorem 5.2.5 The stable homotopy groups define a collection of functors $\pi_n^s : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ as follows.

- For X a pointed CW complex, $\pi_n^s(X)$ is the n th stable homotopy group of X
- For $f : X \rightarrow Y$ a map,

$$\pi_n^s(f) : \pi_n^s(X) \rightarrow \pi_n^s(Y)$$

is the image of f under the canonical map $[X, Y] \rightarrow [X, Y]^s$

Theorem 5.2.6 The stable homotopy functors $\pi_n^s : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ for each $n \in \mathbb{N}$ defines a reduced homology theory.

This is untrue for the unstable homotopy groups. In particular, the fundamental group is not necessarily abelian.

6 Bonus?

6.1 Postnikov Towers

Definition 6.1.1 (Postnikov Towers) Let X be a path connected space. A Postnikov tower is the following commutative diagram

$$\begin{array}{ccccccc}
 & & X & & & & \\
 & & \downarrow & \searrow & \searrow & \searrow & \\
 \cdots & \longrightarrow & X_n & \xrightarrow{p_n} & X_{n-1} & \longrightarrow & \cdots \longrightarrow X_2 \xrightarrow{p_2} X_1 \xrightarrow{p_1} *
 \end{array}$$

such that the following are true.

- The maps $X \rightarrow X_n$ for each $n \in \mathbb{N}$ induces isomorphisms $\pi_i(X) \cong \pi_i(X_n)$ for $i \leq n$.
- $\pi_i(X_n)$ for $i > n$.
- Each $p_n : X_n \rightarrow X_{n-1}$ for $n \in \mathbb{N}$ is a fibration with fiber $K(\pi_n(X), n)$.

Theorem 6.1.2 Suppose that there is an inverse system of spaces

$$\begin{array}{ccccccc}
 & & \lim_{n \rightarrow \infty} X_n & & & & \\
 & & \downarrow & \searrow & \searrow & \searrow & \\
 \cdots & \longrightarrow & X_n & \xrightarrow{p_n} & X_{n-1} & \longrightarrow & \cdots \longrightarrow X_2 \xrightarrow{p_2} X_1 \xrightarrow{p_1} *
 \end{array}$$

The functor π_i for $i \in \mathbb{N}$ induces a cone in **Grp**. By definition of $\lim_{\leftarrow} \pi_i(X_n)$, there is a unique map

$$\lambda : \pi_i \left(\lim_{\leftarrow} X_n \right) \rightarrow \lim_{\leftarrow} \pi_i(X_n)$$

Then the following are true regarding λ .

- λ is surjective
- λ is injective if the maps $\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1})$ are surjective for sufficient large n .

Proposition 6.1.3 Let X be a connected CW complex. Then there exists a Postnikov tower for X .

Proposition 6.1.4 Let X be a connected CW complex. Choose a Postnikov tower of X . Then there is a weak homotopy equivalence

$$X \simeq \lim_{\leftarrow} X_n$$

so that X is a CW approximation of $\lim_{\leftarrow} X_n$.

7 Kunnetth Theorem for Homology

7.1 The Universal Coefficient Theorem for Homology

Proposition 7.1.1 Let (X, A) be a pair of space. Let T be an abelian group. Then there exists a natural map $h : H_n(X, A) \otimes T \rightarrow H_n(X, A; T)$ such that $\text{coker}(h) \cong \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A), T)$ and a split exact sequence (that is not natural) of the form

$$0 \longrightarrow H_n(X, A) \otimes T \xrightarrow{h} H_n(X, A; T) \longrightarrow \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A), T) \longrightarrow 0$$

for any $n \in \mathbb{N}$. In particular, split exactness implies that there is an isomorphism

$$H_n(X, A; T) \cong H_n(X, A) \otimes T \oplus \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A), T)$$

for any $n \in \mathbb{N}$.

7.2 Kunnetth Theorem for Homology

Theorem 7.2.1 (The Eilenberg-Zilbur Theorem)

Let X, Y be spaces. Then $C_\bullet(X) \otimes_{\mathbb{Z}} C_\bullet(Y)$ and $C_\bullet(X \times Y)$ are chain homotopy equivalent, that is natural in X and Y .

Proposition 7.2.2 Let X and Y be CW-complexes. Let R be a principal ideal domain. Then there is a short exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(X; R) \otimes_R H_j(Y; R) \xrightarrow{\times} H_n(X \times Y; R) \longrightarrow \bigoplus_{i+j=n} \text{Tor}_1^R(H_i(X; R), H_{j-1}(Y; R)) \longrightarrow 0$$

induced by the cross product, that is natural in maps $f : X \rightarrow A$ and $g : Y \rightarrow B$. Moreover, this sequence splits.

8 Spectral Sequences in Algebraic Topology

8.1 Spectral Sequences in Topology

Theorem 8.1.1 Let X be a space. Let the following be a sequence

$$\emptyset \subset X_0 \subset X_1 \subset \cdots \subset X$$

of subspaces. Let G be an abelian group. Then the following data

- $A_{p,q} = H_{p+q}(X_p; G)$
- $E_{p,q} = H_{p+q}(X_p, X_{p-1}; G)$
- $i : H_{p+q}(X_p; G) = A_{p,q} \rightarrow H_{p+q}(X_{p+1}; G) = A_{p+1,q-1}$ (degree $(1, -1)$)
- $j : H_{p+q}(X_p; G) = A_{p,q} \rightarrow H_{p+q}(X_p, X_{p-1}; G) = E_{p,q}$ (degree $(0, 0)$)
- $k : H_{p+q}(X_p, X_{p-1}; G) = A_{p,q} \rightarrow H_{p+q-1}(X_{p-1}; G) = A_{p-1,q}$ (degree $(-1, 0)$)

defines an exact couple and hence a spectral sequence with E^1 page given by

$$E_{p,q}^1 = H_{p+q}(X_p, X_{p-1}; G)$$

where the differential $d : E_{p,q}^1 \rightarrow E_{p-1,q}^1$ is given by the composition

$$H_{p+q}(X_p, X_{p-1}; G) \xrightarrow{k} H_{p+q-1}(X_{p-1}; G) \xrightarrow{j} H_{p+q-1}(X_{p-1}, X_{p-2}; G)$$

The E_1 page of such a spectral sequence is given by

$$\begin{array}{ccccccc} \cdots & \longleftarrow & \cdots & \longleftarrow & \cdots & \longleftarrow & \cdots \\ & & H_3(X_1, X_0; G) & \longleftarrow & H_4(X_2, X_1; G) & \longleftarrow & H_4(X_3, X_2; G) \longleftarrow \cdots \\ & & & & & & \\ & & H_2(X_1, X_0; G) & \longleftarrow & H_3(X_2, X_1; G) & \longleftarrow & H_4(X_3, X_2; G) \longleftarrow \cdots \\ & & & & & & \\ & & H_1(X_1, X_0; G) & \longleftarrow & H_2(X_2, X_1; G) & \longleftarrow & H_3(X_3, X_2; G) \longleftarrow \cdots \end{array}$$

Things get interesting when we choose X to be a CW complex and we choose the filtration of X by the skeleton of X . Recall that we have the formula

$$H_{p+q}(X_p, X_{p-1}; G) \cong \begin{cases} C_p^{\text{CW}}(X; G) & \text{if } q = 0 \\ 0 & \text{otherwise} \end{cases}$$

Thus the E^1 page is only left with a chain complex at $q = 0$.

Let us also compute the derived couple of this exact couple or in other words, the E^2 page of the spectral sequence. This is more intuitive than the one thinks about on the definition of the derived couple. The $E_{p,q}^2$ slot is simply the homology of the chain complex at the (p, q) th slot. The direction of the maps of the E^2 page depends not on the choice of spectral sequence at all (In fact, the direction only depends on the page). Now in our case, the homology can be given by a known construct:

$$E_{p,q}^2 = \frac{\ker(d : H_{p+q}(X_p, X_{p-1}; G) \rightarrow H_{p+q-1}(X_{p-1}, X_{p-2}; G))}{\text{im}(d : H_{p+q+1}(X_{p+1}, X_p; G) \rightarrow H_{p+q}(X_p, X_{p-1}; G))} = H_{p+q}^{\text{CW}}(X; G)$$

Since the direction of the maps are now diagonal and when $q \neq 0$ we have $E_{p,q}^2 = 0$, all maps in E^2 are 0 and we are left with

$$H_0^{\text{CW}}(X; G) \quad H_1^{\text{CW}}(X; G) \quad H_2^{\text{CW}}(X; G) \quad H_3^{\text{CW}}(X; G) \quad \cdots$$

Theorem 8.1.2 (Leray-Serre Spectral Sequence) Let $p : E \rightarrow B$ be a Serre fibration with fibre F and path connected B . Suppose that the action of $\pi_1(B)$ on $H_*(F; G)$ is trivial. Then there is a first quadrant homological spectral sequence starting with E^2 and weakly converging to $H_\bullet(E; \mathbb{Z})$. Explicitly, there is a convergence

$$E_{p,q}^2 = H_p(B, H_1(F)) \Rightarrow H_{p+q}(E; \mathbb{Z})$$

9 H-Spaces

9.1 H-Spaces and the Hopf Algebra Structure on Cohomology

Definition 9.1.1 (H-Spaces)

Let X be a space. We say that X is a H-space if there exists a map $\mu : X \times X \rightarrow X$ and a point $e \in X$ such that the maps $x \mapsto \mu(x, e)$ and $x \mapsto \mu(e, x)$ are homotopic to the identity map.

Example 9.1.2

Let G be a topological group. Then G is a H-space.

Example 9.1.3

Let (X, x_0) be a pointed space. Then $\Omega(X, x_0)$ is a H-space.

Lemma 9.1.4

Let X be a H-space. Let Y be a space. Then $[Y, X]$ has multiplication and identity.

Proposition 9.1.5

Let X be a H-space with multiplication given by μ . Let R be a commutative ring. Suppose that the following are true.

- $H^0(X; R) \cong R$.
- $H^n(X; R)$ is a finitely generated free R -module for each $n \in \mathbb{N}$.

Then $H^*(X; R)$ is a Hopf algebra whose comultiplication is induced by μ^* .

9.2 Homotopy Associative H-Space and the Pontryagin Product

Definition 9.2.1 (Homotopy Associative H-Space)

Let X be a H-space. We say that X is homotopy associative if the two maps

$$(x, y, z) \mapsto \mu(x, \mu(y, z)) \quad \text{and} \quad (x, y, z) \mapsto \mu(\mu(x, y), z)$$

are homotopic.

Lemma 9.2.2

Let X be a space. Then X is a group like H-space if and only if X is a monoid object in $\mathbf{hTop}$.

Lemma 9.2.3

Let X be a homotopy associative H-space. Let Y be a space. Then $[Y, X]$ is a monoid.

Definition 9.2.4 (Pontryagin Product)

Let X be a homotopy associative H-space. Let R be a commutative ring. Define the Pontryagin product of X to be the composite

$$H_*(X; R) \otimes_R H_*(X; R) \xrightarrow{\times} H_*(X \times X; R) \xrightarrow{\mu_*} H_*(X; R)$$

where the first map is the cross product given by the Kunneth theorem.

Proposition 9.2.5

Let X be a homotopy associative H-space. Then the Pontryagin product gives a ring structure to the direct sum

$$H_*(X; R) = \bigoplus_{n=0}^{\infty} H_n(X; R)$$

9.3 Group Like H-Spaces

Definition 9.3.1 (Group Like H-Spaces)

Let X be a H-space. We say that X is group like if the following are true.

- X is homotopy associative.
- There exists a map $j : X \rightarrow X$ such that $\mu \circ (\text{id}, j)$ and $\mu \circ (j, \text{id})$ are homotopic to the identity map.

Lemma 9.3.2

Let X be a space. Then X is a group like H-space if and only if X is a group object in $\mathbf{hTop}$.

Lemma 9.3.3

Let X be a group like H-space. Let Y be a space. Then $[Y, X]$ is a group.

Proposition 9.3.4

Let X be homotopy associative H-space. Then X is group like if and only if the map $(x, y) \mapsto (x, xy)$ is a homotopy equivalence.

Definition 9.3.5 (Homotopy Commutative H-Space)

Let X be a H-space. We say that X is homotopy commutative if the two maps

$$(x, y) \mapsto \mu(x, y) \quad \text{and} \quad (x, y) \mapsto \mu(y, x)$$

are homotopic.

Lemma 9.3.6

Let X be a space. Then X is a group like H-space if and only if X is an abelian group object in $\mathbf{hTop}$.

Proposition 9.3.7

Let X be a pointed space. Then the following are true.

- ΩX is a group like H-space.
- $\Omega^2 X$ is group like homotopy commutative H-space.

10 The Free Monoid of a Space

10.1 The James Construction

Definition 10.1.1 (James Reduced Product)

Let (X, e) be a pointed space.

- Define $J_n(X) = \frac{X^n}{\sim}$ where $(x_1, \dots, x_{i-1}, e, x_{i+1}, \dots, x_n) \sim (x_1, \dots, x_i, e, x_{i+2}, \dots, x_n)$.
- Define the James reduced product of (X, e) to be

$$J(X) = \operatorname{colim}_{n \in \mathbb{N}} J_n(X)$$

Lemma 10.1.2

Let (X, e) be a CW complex where e is a 0-cell. Then $J(X)$ is a CW complex.

Proposition 10.1.3

Let X be a connected CW complex. Let R be a ring. Suppose that $H_*(X; R)$ is a free R -module. Then there is an isomorphism

$$H_*(J(X); R) \cong T(\tilde{H}_*(X; R))$$

Let (X, x_0) be a pointed space. Recall that the suspension of X is defined to be

$$\Sigma X = \frac{X \times I}{X \times \partial I \cup \{x_0\} \times I}$$

Elements of ΣX look like (x, t) where $x \in X$ and $t \in I$. Since $(x, 0) \sim (x, 1)$ in the quotient, the map $t \mapsto (x, t)$ is a well defined loop in ΣX , hence an element of $\Omega \Sigma X$.

Proposition 10.1.4

Let (X, e) be a pointed connected CW complex where e is a 0-cell. Let $\lambda : J(X) \rightarrow \Omega \Sigma X$ be the map defined by $\lambda(x) = (t \mapsto (x, t))$. Then λ is a homotopy equivalence.

10.2 The Infinite Symmetric Product

Definition 10.2.1 (Infinite Symmetric Product)

Let (X, e) be a pointed space.

- Define $SP_n(X) = \frac{X^n}{S_n}$ the orbit space, where S^n acts on X^n by $\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma(1)}, \dots, x_{\sigma(n)})$.
- Define the infinite symmetric product of X to be

$$SP(X) = \operatorname{colim}_{n \in \mathbb{N}} SP_n(X)$$

where the maps $SP_n(X) \hookrightarrow SP_{n+1}(X)$ is given by inclusion: $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, e)$.

Lemma 10.2.2

Let (X, e) be a pointed space. Then there is a homeomorphism

$$SP(X) \cong \operatorname{colim}_{n \in \mathbb{N}} \frac{J_n(X)}{S_n}$$

where S_n acts on $J_n(X)$ by $\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma(1)}, \dots, x_{\sigma(n)})$.

Theorem 10.2.3 (Dold-Thom Theorem)

Let (X, e) be a pointed space. Then there are isomorphisms

$$\tilde{H}_n(X) \cong \pi_n(SP(X), e)$$

for each $n \in \mathbb{N}$.

Proposition 10.2.4

Let (X, e) be a pointed space. Then we have a factorization

$$\begin{array}{ccc} \pi_n(X, e) & \xrightarrow{\text{incl}_*} & \pi_n(\text{SP}(X), e) \\ & \searrow h_n & \nearrow \cong \\ & \tilde{H}_n(X) & \end{array}$$

where h is the Hurewicz homomorphism.

11 The Super Lie Algebra Structure in Homotopy Groups

11.1 The Samelson Product

Definition 11.1.1 (Samelson Product)

Let X, Y be spaces. Let G be a group like H -space. Let $f : X \rightarrow G$ and $g : Y \rightarrow G$ be maps. Define the Samelson product $[f, g] : X \wedge Y \rightarrow G$ by

$$[f, g] = [-, -] \circ (f \wedge g)$$

where $[-, -]$ is the commutator map on G .

Note that the Samelson product is well defined because the commutator map $[-, -] : G \times G \rightarrow G$ is trivial on $G \wedge G$ and so it descends to a map $G \wedge G \rightarrow G$.

Proposition 11.1.2

Let G be a group like H -space that is 0-connected. Then the following are true.

- The map $\pi_n(G) \times \pi_m(G) \rightarrow \pi_{n+m}(G)$ given by $([f], [g]) \mapsto [f, g]$ is a $\mathbb{Z}$ -bilinear map.
- Let $[f] \in \pi_n(G)$ and $[g] \in \pi_m(G)$. Then we have

$$[f, g] = (-1)^{nm+1}[g, f]$$

Proposition 11.1.3 (The Super Jacobi Identity)

Let G be a group like H -space that is 0-connected. Let $[f] \in \pi_n(G)$, $[g] \in \pi_m(G)$ and $[h] \in \pi_k(G)$. Then we have

$$(-1)^{nk}[f, [g, h]] + (-1)^{nm}[g, [h, f]] + (-1)^{mk}[h, [f, g]] = 0$$

Corollary 11.1.4

Let G be a group like H -space that is 0-connected. Then $\pi_*(G)$ is a super Lie algebra.

11.2 The Whitehead Product

Definition 11.2.1 (The Whitehead Product)

Let X be a pointed space. Define the Whitehead product on X to be the map $[-, -]_{\text{Wh}} : \pi_n(X) \times \pi_m(X) \rightarrow \pi_{n+m-1}(X)$ as follows. For $[f] \in \pi_n(X)$ and $[g] \in \pi_m(X)$, the Whitehead product is given by

$$[f, g]_{\text{Wh}} = (f \vee g) \circ \phi$$

where $\phi : S^{n+m-1} \rightarrow S^n \vee S^m$ is the attaching map of the $(n+m)$ -cell in the CW complex structure of $S^n \times S^m$.

Proposition 11.2.2

Let X be a pointed space. Let τ be the isomorphism $\pi_{n+1}(X) \cong \pi_n(\Omega X)$. Let $[f] \in \pi_{n+1}(X)$ and $[g] \in \pi_{m+1}(X)$. Then we have

$$\tau([f, g]_{\text{Wh}}) = (-1)^{n-1}[\tau([f]), \tau([g])]$$

Proposition 11.2.3

Let X be a pointed space.

- The Whitehead product $[-, -]_{\text{Wh}}$ is a $\mathbb{Z}$ -bilinear map.
- Let $[f] \in \pi_n(X)$ and $[g] \in \pi_m(X)$. Then we have

$$[f, g] = (-1)^{nm}[g, f]$$

Proposition 11.2.4 (The Super Jacobi Identity)

Let X be a pointed space. Let $[f] \in \pi_n(G)$, $[g] \in \pi_m(G)$ and $[h] \in \pi_k(G)$. Then we have

$$(-1)^{(n-1)(k-1)}[f, [g, h]] + (-1)^{(n-1)(m-1)}[g, [h, f]] + (-1)^{(m-1)(k-1)}[h, [f, g]] = 0$$

Corollary 11.2.5

Let X be a pointed space. Then $\pi_{*-1}(X)$ is a super Lie algebra.

11.3 The Hilton Milnor Theorem

Theorem 11.3.1 (Hilton-Milnor Theorem)

Let X, Y be connected spaces. Then there is a weak equivalence

$$\Omega\Sigma X \times \Omega\Sigma \left(\bigvee_{i=0}^{\infty} (Y \wedge X^{\wedge i}) \right) \rightarrow \Omega\Sigma(X \vee Y)$$

Corollary 11.3.2

Let X, Y be connected spaces. Then there is a zigzag of weak equivalence

$$\Omega\Sigma(X \wedge Y) \simeq \Omega\Sigma X \times \Omega\Sigma Y \times \Omega\Sigma \left(\bigvee_{i,j=1}^{\infty} (X^{\wedge i} \wedge Y^{\wedge j}) \right)$$

given by twice applying the Hilton-Milnor theorem.